

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

_____ show ip route ospf _____

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? _____ S0/0/0 _____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? _____ 129 _____

Does R2 show up as an OSPF neighbor on R1? _____ Yes _____

Does R2 show up as an OSPF neighbor on R3? _____ No _____

What does this information tell you?

_____ R3 reaches 192.168.2.0/24 through R1, not directly through R2. This means the R2–R3 OSPF link is still not exchanging OSPF updates, so R3 uses the longer path via R1. The cost 129 comes from two serial-link costs plus the LAN cost. _____

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

_____ R2(config)# router ospf 1

R2(config-router)# no passive -interface s0/0/1 _____

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? _____ S0/0/1 _____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

_____ 65 (One T1 (1.544 Mb/s) serial link (at a cost of 64) plus the R2 Gigabit 0/0 LAN link (at a cost of 1)). _____

Is R2 listed as an OSPF neighbor to R3? _____ Yes _____

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

_____ The default reference-bandwidth is based on older 100 Mb/s links, so it is not accurate enough for faster interfaces like Gigabit Ethernet. Changing it gives more realistic OSPF costs and better path calculation. _____

Step 2: Change the bandwidth for an interface.

- g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

_____ The cost to 192.168.3.0/24: R1 S0/0/1 + R3 G0/0 ($781+1=782$). The cost to 192.168.23.0/30: R1 S0/0/1 and R3 S0/0/1 ($781+64=845$). _____

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

_____ 1,562. Each serial link now has a cost of 781, and the route to the 192.168.23.0/24 network travels over two serial links. $781 + 781 = 1,562$. _____

Step 3: Change the route cost.

- c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

_____ OSPF will choose the route with the least accumulated cost. The route with the lowest accumulated cost is: R1-S0/0/0 + R2-S0/0/1 + R3-G0/0, or $781 + 781 + 1 = 1,563$. This metric is smaller than the accumulated cost of R1-S0/0/1 + R3-G0/0, or $1565 + 1 = 1,566$. _____

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

____ It is important because the router ID uniquely identifies each OSPF router and is used in neighbor relationships and LSAs. On multi-access networks, it also affects the DR/BDR election. If the router ID is chosen from an active physical interface, it may change when that interface goes down, so it is better to set it manually or use a loopback address. _____

2. Why is the DR/BDR election process not a concern in this lab?

____ It is not a concern because the inter-router links in this lab are serial point-to-point links. DR/BDR election is mainly used on multi-access networks such as Ethernet, not on point-to-point serial links. _____

3. Why would you want to set an OSPF interface to passive?

____ A passive interface stops OSPF Hello packets and routing updates from being sent out that interface. This reduces unnecessary routing traffic on LANs that do not connect to other OSPF routers, while the network can still be advertised through OSPF. _____

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

____ OSPF is hierarchical because it is divided into areas. This area-based design reduces routing overhead and limits SPF calculations and LSAs, which makes large networks more scalable. _____

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

____ network 10.0.0.0 0.255.255.255 area 0 _____

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

____ The EIGRP route will be used. When routes come from different routing protocols, the router compares administrative distance, not the protocol metric. EIGRP has the lowest administrative distance, so it is preferred over OSPF and RIP. OSPF's default administrative distance is 110, and the lab also shows that value in the protocol output. _____